

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Trigonometric Equation

(English Medium)

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TRIGONOMETRIC EQUATION

1. TRIGONOMETRIC EQUATION :

An equation involving one or more trigonometrical ratios of unknown angles is called a trigonometrical equation.

2. SOLUTION OF TRIGONOMETRIC EQUATION :

A value of the unknown angle which satisfies the given equation is called a solution of the trigonometric equation.

- (a) **Principal solution :-** The solution of the trigonometric equation lying in the interval $[0, 2\pi)$.
- (b) **General solution :-** Since all the trigonometric functions are many one & periodic, hence there are infinite values of θ for which trigonometric functions have the same value. All such possible values of θ for which the given trigonometric function is satisfied is given by a general formula. Such a general formula is called general solution of trigonometric equation.
- (c) **Particular solution :-** The solution of the trigonometric equation lying in the given interval.

3. GENERAL SOLUTIONS OF SOME TRIGONOMETRIC EQUATIONS (TO BE REMEMBERED) :

- (a) If $\sin \theta = 0$, then $\theta = n\pi$, $n \in \mathbb{I}$ (set of integers)
- (b) If $\cos \theta = 0$, then $\theta = (2n+1)\frac{\pi}{2}$, $n \in \mathbb{I}$
- (c) If $\tan \theta = 0$, then $\theta = n\pi$, $n \in \mathbb{I}$
- (d) If $\sin \theta = \sin \alpha$, then $\theta = n\pi + (-1)^n \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $n \in \mathbb{I}$
- (e) If $\cos \theta = \cos \alpha$, then $\theta = 2n\pi \pm \alpha$, $n \in \mathbb{I}$, $\alpha \in [0, \pi]$
- (f) If $\tan \theta = \tan \alpha$, then $\theta = n\pi + \alpha$, $n \in \mathbb{I}$, $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (g) If $\sin \theta = 1$, then $\theta = 2n\pi + \frac{\pi}{2} = (4n+1)\frac{\pi}{2}$, $n \in \mathbb{I}$
- (h) If $\cos \theta = 1$ then $\theta = 2n\pi$, $n \in \mathbb{I}$
- (i) If $\sin^2 \theta = \sin^2 \alpha$ or $\cos^2 \theta = \cos^2 \alpha$ or $\tan^2 \theta = \tan^2 \alpha$, then $\theta = n\pi \pm \alpha$, $n \in \mathbb{I}$
- (j) For $n \in \mathbb{I}$, $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$, $n \in \mathbb{I}$
 $\sin(n\pi + \theta) = (-1)^n \sin \theta$ and $\cos(n\pi + \theta) = (-1)^n \cos \theta$
- (k) $\cos n\pi = (-1)^n$, $n \in \mathbb{I}$

If n is an odd integer, then $\sin \frac{n\pi}{2} = (-1)^{\frac{n-1}{2}}$, $\cos \frac{n\pi}{2} = 0$,

$$\sin\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n-1}{2}} \cos \theta$$

$$\cos\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n+1}{2}} \sin \theta$$

Illustration 1 : Find the set of values of x for which $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1$.

Solution : We have, $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1 \Rightarrow \tan(3x - 2x) = 1 \Rightarrow \tan x = 1$
 $\Rightarrow \tan x = \tan \frac{\pi}{4} \Rightarrow x = n\pi + \frac{\pi}{4}, n \in I$ {using $\tan \theta = \tan \alpha \Leftrightarrow \theta = n\pi + \alpha$ }
 But for this value of x , $\tan 2x$ is not defined.
 Hence the solution set for x is ϕ .

Ans.

Do yourself-1 :

1. Find general solutions of the following equations :

(a) $\sin \theta = \frac{1}{2}$

(b) $\cos\left(\frac{3\theta}{2}\right) = 0$

(c) $\tan\left(\frac{3\theta}{4}\right) = 0$

(d) $\cos^2 2\theta = 1$

(e) $\sqrt{3} \sec 2\theta = 2$

(f) $\operatorname{cosec}\left(\frac{\theta}{2}\right) = -1$

2. $\cos 15x = \sin 5x$ if

(A) $x = -\frac{\pi}{20} + \frac{n\pi}{5}, n \in I$

(B) $x = \frac{\pi}{40} + \frac{n\pi}{10}, n \in I$

(C) $x = \frac{3\pi}{20} + \frac{n\pi}{5}, n \in I$

(D) $x = -\frac{3\pi}{40} + \frac{n\pi}{10}, n \in I$

3. $\tan(p\pi/4) = \cot(q\pi/4)$ if :

(A) $p + q = 0$

(B) $p + q = 2n + 1$

(C) $p + q = 2n$

(D) $p + q = 2(2n + 1)$

4. Solve $\tan 2\theta = \tan \frac{2}{\theta}$.

5. What are the most general values θ of which satisfy the equations :

(i) $\sin \theta = \frac{1}{\sqrt{2}}$

(ii) $\tan(x-1) = \sqrt{3}$

(iii) $\tan \theta = -1$

(iv) $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$

(v) $2\cot^2 \theta = \operatorname{cosec}^2 \theta$

6. Solve :

(i) $\sin 9\theta = \sin \theta$

(ii) $\cot \theta + \tan \theta = 2\operatorname{cosec} \theta$

(iii) $\sin 2\theta = \cos 3\theta$

(iv) $\cot \theta = \tan 8\theta$

(v) $\cot \theta - \tan \theta = 2$

(vi) $\operatorname{cosec} \theta = \cot \theta + \sqrt{3}$

(vii) $\tan 2\theta \tan \theta = 1$

(viii) $\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$

- (a) For equations of the type $\sin \theta = k$ or $\cos \theta = k$, one must check that $|k| \leq 1$.
- (b) Avoid squaring the equations, if possible, because it may lead to extraneous solutions. Reject extra solutions if they do not satisfy the given equation.
- (c) Do not cancel the common variable factor from the two sides of the equations which are in a product because we may lose some solutions.
- (d) The answer should not contain such values of θ , which make any of the terms undefined or infinite.
 - (i) Check that denominator is not zero at any stage while solving equations.
 - (ii) If $\tan \theta$ or $\sec \theta$ is involved in the equations, θ should not be odd multiple of $\frac{\pi}{2}$.
 - (iii) If $\cot \theta$ or $\operatorname{cosec} \theta$ is involved in the equation, θ should not be multiple of π or 0 .

(a) Solving trigonometric equations by factorisation.

$$\therefore (2 \sin x - \cos x)(1 + \cos x) - (1 - \cos^2 x) = 0$$

$$\therefore (1 + \cos x)(2 \sin x - \cos x - 1 + \cos x) = 0$$

$$\therefore (1 + \cos x)(2 \sin x - 1) = 0$$

$$\Rightarrow \cos x = -1 \text{ or } \sin x = \frac{1}{2}$$

$$\Rightarrow \cos x = -1 = \cos \pi \Rightarrow x = 2n\pi + \pi = (2n + 1)\pi, n \in \mathbb{I}$$

$$\text{or } \sin x = \frac{1}{2} = \sin \frac{\pi}{6} \Rightarrow x = k\pi + (-1)^k \frac{\pi}{6}, k \in I$$

(A) $2n\pi \pm \frac{\pi}{3}$ (B) $2n\pi \pm \frac{\pi}{6}$ (C) $n\pi \pm \frac{\pi}{3}$ (D) none of these

Since, $\frac{1}{6} \sin \theta, \cos \theta, \tan \theta$ are in G.P.

$$\Rightarrow \cos^2 \theta = \frac{1}{6} \sin \theta \cdot \tan \theta \Rightarrow 6\cos^3 \theta + \cos^2 \theta - 1 = 0$$

$$\therefore (2\cos \theta - 1)(3 \cos^2 \theta + 2 \cos \theta + 1) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2} \quad (\text{other values of } \cos \theta \text{ are imaginary})$$

$$\Rightarrow \cos \theta = \cos \frac{\pi}{3} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{I}.$$

Ans. (A)

(b) Solving of trigonometric equation by reducing it to a quadratic equation.

e.g. $6 - 10\cos x = 3\sin^2 x$

$$\therefore 6 - 10\cos x = 3 - 3\cos^2 x \Rightarrow 3\cos^2 x - 10\cos x + 3 = 0$$

$$\Rightarrow (3\cos x - 1)(\cos x - 3) = 0 \Rightarrow \cos x = \frac{1}{3} \text{ or } \cos x = 3$$

Since $\cos x = 3$ is not possible as $-1 \leq \cos x \leq 1$

$$\therefore \cos x = \frac{1}{3} = \cos\left(\cos^{-1}\frac{1}{3}\right) \Rightarrow x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right), n \in \mathbb{I}$$

Illustration 3 : Solve $\sin^2 \theta - \cos \theta = \frac{1}{4}$ for θ and write the values of θ in the interval $0 \leq \theta \leq 2\pi$.

Solution : The given equation can be written as

$$1 - \cos^2 \theta - \cos \theta = \frac{1}{4} \Rightarrow \cos^2 \theta + \cos \theta - 3/4 = 0$$

$$\Rightarrow 4\cos^2 \theta + 4\cos \theta - 3 = 0 \Rightarrow (2\cos \theta - 1)(2\cos \theta + 3) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2}, -\frac{3}{2}$$

Since, $\cos \theta = -3/2$ is not possible as $-1 \leq \cos \theta \leq 1$

$$\therefore \cos \theta = \frac{1}{2} \Rightarrow \cos \theta = \cos \frac{\pi}{3} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{I}$$

For the given interval, $n = 0$ and $n = 1$.

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Ans.

Illustration 4 : Find the number of solutions of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$.

Solution : Here, $\tan x + \sec x = 2\cos x \Rightarrow \sin x + 1 = 2\cos^2 x$

$$\Rightarrow 2\sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}, -1$$

But $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$ for which $\tan x + \sec x = 2\cos x$ is not defined.

$$\text{Thus } \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$\Rightarrow$ number of solutions of $\tan x + \sec x = 2\cos x$ is 2.

Ans.

Illustration 5: Solve the equation $5\sin^2 x - 7\sin x \cos x + 16\cos^2 x = 4$

Solution : To solve this equation we use the fundamental formula of trigonometric identities,

$$\sin^2 x + \cos^2 x = 1$$

writing the equation in the form,

$$5\sin^2x - 7\sin x \cdot \cos x + 16\cos^2x = 4(\sin^2x + \cos^2x)$$

$$\Rightarrow \sin^2 x - 7 \sin x \cos x + 12 \cos^2 x = 0$$

dividing by $\cos^2 x$ on both side we get,

$$\tan^2 x - 7 \tan x + 12 = 0$$

Now it can be factorized as :

$$(\tan x - 3)(\tan x - 4) = 0$$

$$\Rightarrow \tan x = 3,4$$

i.e., $\tan x = \tan(\tan^{-1} 3)$ or $\tan x = \tan(\tan^{-1} 4)$

$$\Rightarrow x = n\pi + \tan^{-1} 3 \text{ or } x = n\pi + \tan^{-1} 4, n \in \mathbb{I}.$$

Ans.

Illustration 6: If $x \neq \frac{n\pi}{2}$, $n \in \mathbb{I}$ and $(\cos x)^{\sin^2 x - 3 \sin x + 2} = 1$, then find the general solutions of x .

Solution : As $x \neq \frac{n\pi}{2} \Rightarrow \cos x \neq 0, 1, -1$

So, $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1 \Rightarrow \sin^2 x - 3\sin x + 2 = 0$

$$\therefore (\sin x - 2)(\sin x - 1) = 0 \Rightarrow \sin x = 1, 2$$

where $\sin x = 2$ is not possible and $\sin x = 1$ which is also not possible as $x \neq \frac{n\pi}{2}$

$\therefore$ no general solution is possible.

Ans.

Illustration 7: Solve the equation $\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cdot \cos x$.

Solution : $\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cdot \cos x \Rightarrow (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = \frac{7}{2} \sin x \cdot \cos x$

$$\Rightarrow 1 - \frac{1}{2}(\sin 2x)^2 = \frac{7}{4}(\sin 2x) \Rightarrow 2\sin^2 2x + 7\sin 2x - 4 = 0$$

$$\Rightarrow (2\sin 2x - 1)(\sin 2x + 4) = 0 \Rightarrow \sin 2x = \frac{1}{2} \text{ or } \sin 2x = -4 \text{ (which is not possible)}$$

$$\Rightarrow 2x = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{I}$$

$$\text{i.e., } x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}, n \in I$$

Ans.

Do yourself-2 :

1. Solve the following equations :

(a) $3\sin x + 2\cos^2 x = 0$

(b) $\sec^2 2\alpha = 1 - \tan 2\alpha$

(c) $7\cos^2 \theta + 3\sin^2 \theta = 4$

(d) $4\cos \theta - 3\sec \theta = \tan \theta$

2. Solve the equation : $2\sin^2 \theta + \sin^2 2\theta = 2$ for $\theta \in (-\pi, \pi)$.3. Solve $\cos^3 x + \cos^2 x - 4\cos^2 \frac{x}{2} = 0$ 4. Solve $\cot^2 \theta + 3\operatorname{cosec} \theta + 3 = 0$ 5. The general solution of the equation, $\frac{1 - \sin x + \dots + (-1)^{n-1} \sin^{n-1} x + \dots \infty}{1 + \sin x + \dots + \sin^{n-1} x + \dots \infty} = \frac{1 - \cos 2x}{1 + \cos 2x}$ is

(A) $(-1)^n \pi/3 + n\pi$

(B) $(-1)^n \pi/6 + n\pi$

(C) $(-1)^{n+1} \pi/6 + n\pi$

(D) $(-1)^{n-1} \pi/3 + n\pi$

6. If sum of all solutions of the equation $6 \sin \frac{x}{2} = \sec \frac{x}{2}$ in $[-2\pi, 4\pi]$ is $k\pi$, then k is equal to7. If sum of all possible values of $x \in (0, 2\pi)$ satisfying the equation

$$2 \cos x \cdot \operatorname{cosec} x - 4 \cos x - \operatorname{cosec} x = -2, \text{ is } \frac{k\pi}{4}, \text{ then k is equal to}$$

(A) 9

(B) 12

(C) 16

(D) 32

8. Solve

(i) $\tan^2 \theta - (1 + \sqrt{3}) \tan \theta + \sqrt{3} = 0$

(ii) $4 \cos \theta - 3 \sec \theta = 2 \tan \theta$

(iii) $\tan x \cdot \tan \left(x + \frac{\pi}{3} \right) \cdot \tan \left(x + \frac{2\pi}{3} \right) = \sqrt{3}$

(c) Solving trigonometric equations by introducing an auxilliary argument.Consider, $a \sin \theta + b \cos \theta = c$ (i)

$$\therefore \frac{a}{\sqrt{a^2 + b^2}} \sin \theta + \frac{b}{\sqrt{a^2 + b^2}} \cos \theta = \frac{c}{\sqrt{a^2 + b^2}}$$

equation (i) has a solution only if $|c| \leq \sqrt{a^2 + b^2}$

$$\text{let } \frac{a}{\sqrt{a^2 + b^2}} = \cos \phi, \quad \frac{b}{\sqrt{a^2 + b^2}} = \sin \phi \quad \& \quad \phi = \tan^{-1} \frac{b}{a}$$

by introducing this auxilliary argument ϕ , equation (i) reduces to

$$\sin (\theta + \phi) = \frac{c}{\sqrt{a^2 + b^2}} \quad \text{Now this equation can be solved easily.}$$

Illustration 8 : Find the number of distinct solutions of $\sec x + \tan x = \sqrt{3}$, where $0 \leq x \leq 3\pi$.

Solution : Here, $\sec x + \tan x = \sqrt{3} \Rightarrow 1 + \sin x = \sqrt{3} \cos x$

$$\text{or } \sqrt{3} \cos x - \sin x = 1$$

dividing both sides by $\sqrt{a^2 + b^2}$ i.e. $\sqrt{4} = 2$, we get

$$\Rightarrow \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \frac{1}{2}$$

$$\Rightarrow \cos \frac{\pi}{6} \cos x - \sin \frac{\pi}{6} \sin x = \frac{1}{2} \Rightarrow \cos \left(x + \frac{\pi}{6} \right) = \frac{1}{2}$$

As $0 \leq x \leq 3\pi$

$$\frac{\pi}{6} \leq x + \frac{\pi}{6} \leq 3\pi + \frac{\pi}{6}$$

$$\Rightarrow x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow x = \frac{\pi}{6}, \frac{3\pi}{2}, \frac{13\pi}{6}$$

But at $x = \frac{3\pi}{2}$, $\tan x$ and $\sec x$ is not defined.

$\therefore$ Total number of solutions are 2.

Ans.

Illustration 9 : Prove that the equation $k \cos x - 3 \sin x = k + 1$ possess a solution iff $k \in (-\infty, 4]$.

Solution : Here, $k \cos x - 3 \sin x = k + 1$, could be re-written as :

$$\frac{k}{\sqrt{k^2 + 9}} \cos x - \frac{3}{\sqrt{k^2 + 9}} \sin x = \frac{k+1}{\sqrt{k^2 + 9}}$$

$$\text{or } \cos(x + \phi) = \frac{k+1}{\sqrt{k^2 + 9}}, \text{ where } \tan \phi = \frac{3}{k}$$

which possess a solution only if $-1 \leq \frac{k+1}{\sqrt{k^2 + 9}} \leq 1$

$$\text{i.e., } \left| \frac{k+1}{\sqrt{k^2 + 9}} \right| \leq 1$$

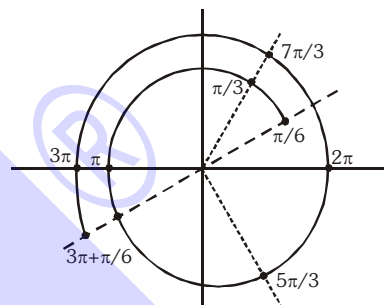
$$\text{i.e., } (k+1)^2 \leq k^2 + 9$$

$$\text{i.e., } k^2 + 2k + 1 \leq k^2 + 9$$

$$\text{or } k \leq 4$$

$\Rightarrow$ The interval of k for which the equation $(k \cos x - 3 \sin x = k + 1)$ has a solution is $(-\infty, 4]$.

Ans.



Do yourself-3 :**1** Solve the following equations :

(a) $\sin x + \sqrt{2} = \cos x$ (b) $\operatorname{cosec} \theta = 1 + \cot \theta$

2. Solve : $\cos \theta + \sin \theta = \cos 2\theta + \sin 2\theta$ **3.** Solve : $8 \sin x = \frac{\sqrt{3}}{\cos x} + \frac{1}{\sin x}$ **4.** Solve

(i) $\sqrt{3} \sin \theta - \cos \theta = \sqrt{2}$ (ii) $5 \sin \theta + 2 \cos \theta = 5$

(d) Solving trigonometric equations by transforming sum of trigonometric functions into product.

e.g. $\cos 3x + \sin 2x - \sin 4x = 0$

$$\cos 3x - 2 \sin x \cos 3x = 0$$

$$\Rightarrow (\cos 3x)(1 - 2 \sin x) = 0$$

$$\Rightarrow \cos 3x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow \cos 3x = 0 = \cos \frac{\pi}{2} \quad \text{or} \quad \sin x = \frac{1}{2} = \sin \frac{\pi}{6}$$

$$\Rightarrow 3x = 2n\pi \pm \frac{\pi}{2} \quad \text{or} \quad x = m\pi + (-1)^m \frac{\pi}{6}$$

$$\Rightarrow x = \frac{2n\pi}{3} \pm \frac{\pi}{6} \quad \text{or} \quad x = m\pi + (-1)^m \frac{\pi}{6}; (n, m \in \mathbb{I})$$

Illustration 10 : Solve : $\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$ **Solution :** We have $\cos \theta + \cos 7\theta + \cos 3\theta + \cos 5\theta = 0$

$$\Rightarrow 2 \cos 4\theta \cos 3\theta + 2 \cos 4\theta \cos \theta = 0 \Rightarrow \cos 4\theta (\cos 3\theta + \cos \theta) = 0$$

$$\Rightarrow \cos 4\theta (2 \cos 2\theta \cos \theta) = 0$$

$$\Rightarrow \text{Either } \cos \theta = 0 \Rightarrow \theta = (2n_1 + 1) \frac{\pi}{2}, n_1 \in \mathbb{I}$$

$$\text{or } \cos 2\theta = 0 \Rightarrow \theta = (2n_2 + 1) \frac{\pi}{4}, n_2 \in \mathbb{I}$$

$$\text{or } \cos 4\theta = 0 \Rightarrow \theta = (2n_3 + 1) \frac{\pi}{8}, n_3 \in \mathbb{I}$$

Ans.

(e) Solving trigonometric equations by transforming a product into sum.

e.g. $\sin 5x \cdot \cos 3x = \sin 6x \cdot \cos 2x$

$$\sin 8x + \sin 2x = \sin 8x + \sin 4x$$

$$\therefore 2\sin 2x \cdot \cos 2x - \sin 2x = 0$$

$$\Rightarrow \sin 2x(2 \cos 2x - 1) = 0$$

$$\Rightarrow \sin 2x = 0 \quad \text{or} \quad \cos 2x = \frac{1}{2}$$

$$\Rightarrow \sin 2x = 0 = \sin 0 \quad \text{or} \quad \cos 2x = \frac{1}{2} = \cos \frac{\pi}{3}$$

$$\Rightarrow 2x = n\pi + (-1)^n \times 0, n \in I \quad \text{or} \quad 2x = 2m\pi \pm \frac{\pi}{3}, m \in I$$

$$\Rightarrow x = \frac{n\pi}{2}, n \in I \quad \text{or} \quad x = m\pi \pm \frac{\pi}{6}, m \in I$$

Illustration 11 : Solve : $\cos \theta \cos 2\theta \cos 3\theta = \frac{1}{4}$; where $0 \leq \theta \leq \pi$.

Solution : $\frac{1}{2} (2\cos \theta \cos 3\theta) \cos 2\theta = \frac{1}{4} \Rightarrow (\cos 2\theta + \cos 4\theta) \cos 2\theta = \frac{1}{2}$

$$\Rightarrow \frac{1}{2} [2\cos^2 2\theta + 2\cos 4\theta \cos 2\theta] = \frac{1}{2} \Rightarrow 1 + \cos 4\theta + 2\cos 4\theta \cos 2\theta = 1$$

$$\therefore \cos 4\theta (1 + 2\cos 2\theta) = 0$$

$$\cos 4\theta = 0 \quad \text{or} \quad (1 + 2\cos 2\theta) = 0$$

Now from the first equation : $2\cos 4\theta = 0 = \cos(\pi/2)$

$$\therefore 4\theta = \left(n + \frac{1}{2}\right)\pi \Rightarrow \theta = (2n + 1)\frac{\pi}{8}, n \in I$$

for $n = 0, \theta = \frac{\pi}{8}; n = 1, \theta = \frac{3\pi}{8}; n = 2, \theta = \frac{5\pi}{8}; n = 3, \theta = \frac{7\pi}{8} \quad (\because 0 \leq \theta \leq \pi)$

and from the second equation :

$$\cos 2\theta = -\frac{1}{2} = -\cos(\pi/3) = \cos(\pi - \pi/3) = \cos(2\pi/3)$$

$$\therefore 2\theta = 2k\pi \pm 2\pi/3 \therefore \theta = k\pi \pm \pi/3, k \in I$$

again for $k = 0, \theta = \frac{\pi}{3}; k = 1, \theta = \frac{2\pi}{3} \quad (\because 0 \leq \theta \leq \pi)$

$$\therefore \theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$$

Ans.

Illustration 12 : The general solution of the trigonometric equation $\sin x \cos 2x + \sin 2x \cos 5x = \sin 3x \cos 5x$, is

- (A) $\frac{n\pi}{3}$ (B) $\frac{2n\pi}{9}$ (C) $2n\pi$ (D) $\frac{n\pi}{3} \cup \frac{2n\pi}{9}$

(where $n \in \mathbb{I}$)

Solution : $2 \sin x \cos 2x + 2 \sin 2x \cos 5x = 2 \sin 3x \cos 5x$

$$\sin 3x - \sin x + \sin 7x - \sin 3x = \sin 8x - \sin 2x$$

$$\sin 7x - \sin x = \sin 8x - \sin 2x$$

$$2 \cos 4x \sin 3x = 2 \cos 5x \sin 3x$$

$$2 \sin 3x [\cos 5x - \cos 4x] = 0$$

$$\sin 3x = 0 \Rightarrow x = \frac{n\pi}{3}$$

$$\text{if } \cos 5x - \cos 4x = 0$$

$$2 \sin \frac{9x}{2} \sin \frac{x}{2} = 0$$

$$\therefore x = \frac{2n\pi}{9} \text{ or } 2n\pi$$

$$\text{Hence general solution is } \frac{n\pi}{3} \cup \frac{2n\pi}{9}, n \in \mathbb{I}$$

Do yourself-4 :

- Solve : $4 \sin \theta \sin 2\theta \sin 4\theta = \sin 3\theta$.
- Solve for x : $\sin x + \sin 3x + \sin 5x = 0$.
- Solve : $\sin 2x + 5 \sin x + 1 + 5 \cos x = 0$
- Solve : $3 \cos x + 3 \sin x + \sin 3x - \cos 3x = 0$
- Solve : $(1 - \sin 2x)(\cos x - \sin x) = 1 - 2 \sin^2 x$.
- The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a solution is
(A) 2 (B) 3 (C) 4 (D) 5
- Solve
 - $\cos \theta + \sin \theta = \cos 2\theta + \sin 2\theta$
 - $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$
 - $\sin^2 n\theta - \sin^2(n-1)\theta = \sin^2 \theta$, where n is constant and $n \neq 0, 1$

(f) Solving equations by a change of variable :

- (i) Equations of the form $P(\sin x \pm \cos x, \sin x \cdot \cos x) = 0$, where $P(y, z)$ is a polynomial, can be solved by the substitution :

$$\cos x \pm \sin x = t \Rightarrow 1 \pm 2 \sin x \cdot \cos x = t^2.$$

Solution :

$$\begin{aligned} \text{put } \sin x + \cos x &= t \\ \Rightarrow \sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x &= t^2 \\ \Rightarrow 2 \sin x \cos x &= t^2 - 1 \quad (\because \sin^2 x + \cos^2 x = 1) \\ \Rightarrow \sin x \cdot \cos x &= \left(\frac{t^2 - 1}{2} \right) \end{aligned}$$

Substituting above result in given equation, we get :

$$\begin{aligned} t &= 1 + \frac{t^2 - 1}{2} \\ \Rightarrow 2t &= t^2 + 1 \Rightarrow t^2 - 2t + 1 = 0 \\ \Rightarrow (t - 1)^2 &= 0 \quad \Rightarrow \quad t = 1 \\ \Rightarrow \sin x + \cos x &= 1 \end{aligned}$$

Dividing both sides by $\sqrt{1^2 + 1^2}$ i.e. $\sqrt{2}$, we get

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{1}{\sqrt{2}} \Rightarrow \cos x \cos \frac{\pi}{4} + \sin x \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$
$$\Rightarrow \cos \left(x - \frac{\pi}{4} \right) = \cos \frac{\pi}{4} \Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$$
$$\Rightarrow x = 2n\pi \text{ or } x = 2n\pi + \frac{\pi}{2} = (4n + 1) \frac{\pi}{2}, n \in \mathbb{I}$$

(ii) Equations of the form $a \sin x + b \cos x + d = 0$, where a , b & d are real numbers can be solved by changing $\sin x$ & $\cos x$ into their corresponding tangent of half the angle.

Illustration 14: Solve : $3 \cos x + 4 \sin x = 5$

Solution : $\Rightarrow 3\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right)+4\left(\frac{2 \tan x/2}{1+\tan^2 x/2}\right)=5$

$$\Rightarrow \frac{3-3\tan^2 \frac{x}{2}}{1+\tan^2 \frac{x}{2}} + \frac{8\tan \frac{x}{2}}{1+\tan^2 \frac{x}{2}} = 5$$

$$\Rightarrow 3 - 3\tan^2 \frac{x}{2} + 8\tan \frac{x}{2} = 5 + 5\tan^2 \frac{x}{2} \Rightarrow 8\tan^2 \frac{x}{2} - 8\tan \frac{x}{2} + 2 = 0$$

$$\Rightarrow 4 \tan^2 \frac{x}{2} - 4 \tan \frac{x}{2} + 1 = 0 \quad \Rightarrow \quad \left(2 \tan \frac{x}{2} - 1 \right)^2 = 0$$

$$\Rightarrow 2 \tan \frac{x}{2} - 1 = 0 \Rightarrow \tan \frac{x}{2} = \frac{1}{2} = \tan \left(\tan^{-1} \frac{1}{2} \right)$$

$$\Rightarrow \frac{x}{2} = n\pi + \tan^{-1}\left(\frac{1}{2}\right), n \in \mathbb{I} \Rightarrow x = 2n\pi + 2\tan^{-1}\frac{1}{2}, n \in \mathbb{I}$$

(g) Solving trigonometric equations with the use of the boundness of the functions involved.

Illustration 17: The number of solution(s) of $2\cos^2\left(\frac{x}{2}\right)\sin^2x = x^2 + \frac{1}{x^2}$, $0 \leq x \leq \pi/2$, is/are -

- (A) 0 (B) 1 (C) infinite (D) none of these

Solution : Let $y = 2\cos^2\left(\frac{x}{2}\right)\sin^2x = x^2 + \frac{1}{x^2} \Rightarrow y = (1 + \cos x)\sin^2x$ and $y = x^2 + \frac{1}{x^2}$

when $y = (1 + \cos x)\sin^2 x = (\text{a number} < 2)(\text{a number} \leq 1) \Rightarrow y < 2$ (i)

and when $y = x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2 \geq 2 \Rightarrow y \geq 2$ (ii)

No value of y can be obtained satisfying (i) and (ii), simultaneously

$\Rightarrow$ No real solution of the equation exists.

Ans. (A)

Note: If L.H.S. of the given trigonometric equation is always less than or equal to k and RHS is always greater than k , then no solution exists. If both the sides are equal to k for same value of θ , then solution exists and if they are equal for different values of θ , then solution does not exist.

Illustration 18 : The number of ordered pairs (x, y) satisfying $|x| + |y| = 3$ and $\sin\left(\frac{\pi x^2}{3}\right) = 1$ is less than or equal to

- (A) 7 (B) 8 (C) 9 (D) 10

Solution : $\sin\left(\frac{\pi}{3}x^2\right) = 1$

$$\frac{\pi}{3}x^2 = 2n\pi + \frac{\pi}{2}; n \in \mathbb{I}$$

$$\frac{\pi}{3}x^2 = (4n + 1) \frac{\pi}{2}$$

$$x^2 = \frac{3}{2}(4n + 1); n \in \mathbb{I}$$

only $n = 0$ and $n = 1$ is possible.

$$x^2 = \frac{3}{2} \quad \text{or} \quad x^2 = \frac{15}{2}$$

$$\therefore x = \pm \sqrt{\frac{3}{2}} \quad \text{or} \quad x = \pm \sqrt{\frac{15}{2}}.$$

for each value of x will get 2 values of y , Hence 8 ordered pairs.

Do yourself-5 :

- If $x^2 - 4x + 5 - \sin y = 0$, $y \in [0, 2\pi)$, then -
 (A) $x = 1, y = 0$ (B) $x = 1, y = \pi/2$ (C) $x = 2, y = 0$ (D) $x = 2, y = \pi/2$
- If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $y > 0$, $x \in [0, \pi]$, then find the least positive value of x satisfying the given condition.
- Solve $\sin 3x + \cos 2x = -2$
- Solve $\sqrt{3 \sin 5x - \cos^2 x - 3} = 1 - \sin x$
- The number of real solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is -
 (A) 0 (B) 1 (C) 2 (D) infinitely many
- If $\sin^3 \alpha + \cos^3 \beta + 6 \sin \alpha \cdot \cos \beta = 8$, where $\alpha, \beta \in [0, 2\pi]$ then number of ordered pairs (α, β) is equal to
 (A) 0 (B) 1 (C) 2 (D) 3

6. TRIGONOMETRIC INEQUALITIES :

There is no general rule to solve trigonometric inequations and the same rules of algebra are valid provided the domain and range of trigonometric functions should be kept in mind.

Illustration 19 : Find the solution set of inequality $\sin x > 1/2$.

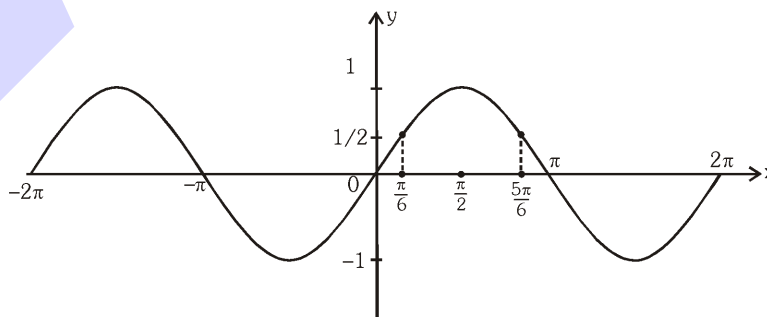
Solution : When $\sin x = \frac{1}{2}$, the two values of x between 0 and 2π are $\pi/6$ and $5\pi/6$.

From the graph of $y = \sin x$, it is obvious that between 0 and 2π ,

$$\sin x > \frac{1}{2} \text{ for } \pi/6 < x < 5\pi/6$$

Hence, $\sin x > 1/2$

$$\Rightarrow 2n\pi + \pi/6 < x < 2n\pi + 5\pi/6, n \in \mathbb{I}$$



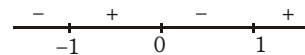
Thus, the required solution set is $\bigcup_{n \in \mathbb{I}} \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6} \right)$

Ans.

Illustration 20 : Find the values of α lying between 0 and π for which the inequality : $\tan \alpha > \tan^3 \alpha$ is valid.

Solution : We have : $\tan \alpha - \tan^3 \alpha > 0 \Rightarrow \tan \alpha (1 - \tan^2 \alpha) > 0$

$$\Rightarrow (\tan \alpha)(\tan \alpha + 1)(\tan \alpha - 1) < 0$$



So $\tan \alpha < -1, 0 < \tan \alpha < 1$

$$\therefore \text{ Given inequality holds for } \alpha \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$$

Ans.

Do yourself - 6 :

1. Find the solution set of the inequality : $\cos x \geq -1/2$.
2. Find the values of x in the interval $[0, 2\pi]$ for which $4\sin^2 x - 8\sin x + 3 \leq 0$.
3. Solve $\tan^2 x \leq 1$
4. Solve $2\sin^2 x - \sin x - 1 > 0$
5. Solve $\sqrt{\sqrt{3} \cot \theta} < 1$

Miscellaneous Illustration :

Illustration 21 : Solve the following equation : $\tan^2 \theta + \sec^2 \theta + 3 = 2(\sqrt{2} \sec \theta + \tan \theta)$

Solution : We have $\tan^2 \theta + \sec^2 \theta + 3 = 2\sqrt{2} \sec \theta + 2 \tan \theta$

$$\Rightarrow \tan^2 \theta - 2 \tan \theta + \sec^2 \theta - 2\sqrt{2} \sec \theta + 3 = 0$$

$$\Rightarrow \tan^2 \theta + 1 - 2 \tan \theta + \sec^2 \theta - 2\sqrt{2} \sec \theta + 2 = 0$$

$$\Rightarrow (\tan \theta - 1)^2 + (\sec \theta - \sqrt{2})^2 = 0 \quad \Rightarrow \quad \tan \theta = 1 \text{ and } \sec \theta = \sqrt{2}$$

As the periodicity of $\tan \theta$ and $\sec \theta$ are not same, we get

$$\theta = 2n\pi + \frac{\pi}{4}, n \in \mathbb{I}$$

Ans.

Illustration 22 : Find the solution set of equation $5^{(1 + \log_5 \cos x)} = 5/2$.

Solution : Taking log to base 5 on both sides in given equation :

$$(1 + \log_5 \cos x) \cdot \log_5 5 = \log_5 (5/2) \Rightarrow \log_5 5 + \log_5 \cos x = \log_5 5 - \log_5 2$$

$$\Rightarrow \log_5 \cos x = -\log_5 2 \Rightarrow \cos x = 1/2 \Rightarrow x = 2n\pi \pm \pi/3, n \in \mathbb{I}$$

Ans.

Illustration 23 : If the set of all values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4 \sin x + \sqrt{2}| < \sqrt{6}$ is $\left(\frac{a\pi}{24}, \frac{b\pi}{24}\right)$ then find the value of $\left|\frac{a-b}{3}\right|$.

Solution :

$$|4 \sin x + \sqrt{2}| < \sqrt{6}$$

$$\Rightarrow -\sqrt{6} < 4 \sin x + \sqrt{2} < \sqrt{6} \Rightarrow -\sqrt{6} - \sqrt{2} < 4 \sin x < \sqrt{6} - \sqrt{2}$$

$$\Rightarrow \frac{-(\sqrt{6} + \sqrt{2})}{4} < \sin x < \frac{\sqrt{6} - \sqrt{2}}{4} \Rightarrow -\frac{5\pi}{12} < x < \frac{\pi}{12} \text{ for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Comparing with $\frac{a\pi}{24} < x < \frac{b\pi}{24}$, we get, $a = -10, b = 2$

$$\therefore \left|\frac{a-b}{3}\right| = \left|\frac{-10-2}{3}\right| = 4$$

Ans.

Illustration 24 : The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is -

- (A) 0 (B) 5 (C) 6 (D) 10

Solution :

$$3 \sin^2 x - 7 \sin x + 2 = 0$$

$$\Rightarrow (3 \sin x - 1)(\sin x - 2) = 0$$

$$\therefore \sin x \neq 2$$

$$\Rightarrow \sin x = \frac{1}{3} = \sin \alpha \text{ (say)}$$

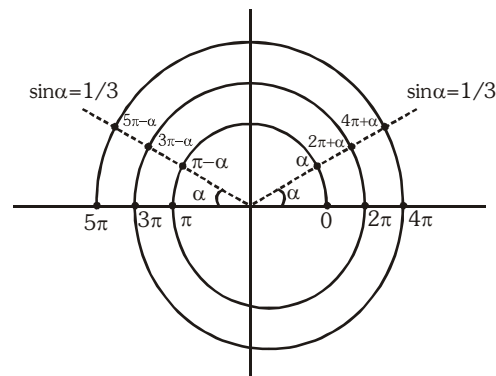
where α is the least positive value of x

such that $\sin \alpha = \frac{1}{3}$.

Clearly $0 < \alpha < \frac{\pi}{2}$. We get the solution,

$$x = \alpha, \pi - \alpha, 2\pi + \alpha, 3\pi - \alpha, 4\pi + \alpha \text{ and } 5\pi - \alpha.$$

Hence total six values in $[0, 5\pi]$



Ans. (C)